Chapter 10

Number Theory

Well-Ordering Principle: Every non-empty subset of non-negative integers set has a minimum element.

10.1 Definition

Definition 10.1.1. Let $a, b \in \mathbb{Z}$ be integers.
We said to be b divides a if: $a = bx$ for some $x \in \mathbb{Z}$. Denote $b \mid a$.

10.2 The Division Algorithm

Theorem 10.2.1. Let $a, b \in \mathbb{Z}$ be integers with $b > 0$. Then, there exists unique integers $q, r \in \mathbb{Z}$ such that

$$a = b \cdot q + r \quad (0 \leq r < b)$$

Proof. Let $S \stackrel{\text{def}}{=} \{a - bx \mid x \in \mathbb{Z},\ a - bx \geq 0\}$. Then, S is non-empty because $a - b \cdot (-|a|) = a + |a|b \geq a + |a| \geq 0$.
Thus, Well-Ordering Principle gives that there exists a minimum element of S. Put $r = \min S$.
(That is, for any $k \in S$, $r \leq k$.) Since $r \in S$, there exists $q \in \mathbb{Z}$ such that $r = a - bq \geq 0$.
To show $r < b$, we will use Contradiction. Suppose that $r \geq b$. Then,

$$a - b \cdot (q + 1) = a - bq - b = r - b \geq 0$$

Thus, $a - b(q + 1) \in S$, but this is contradiction with $r = a - bq$ is a minimum element of S.
To show the Uniqueness, there exist $q, q', r, r' \in \mathbb{Z}$ such that

$$a = bq + r = bq' + r' \quad (0 \leq r, r' < b)$$

Then, $b|q - q'| = |r - r'|$, thus must be $q = q'$ and $r = r'$. $\square$

Corollary 10.2.2. Let $a, b \in \mathbb{Z}$ be integers with $b \neq 0$. Then, there exists unique integers $q, r \in \mathbb{Z}$ such that

$$a = b \cdot q + r \quad (0 \leq r < |b|)$$

10.3 Bézout's Identity

Definition 10.3.1. Let $a, b \in \mathbb{Z}$ be integers. The integer d is called *The greatset common divisor* of a, b if:

1. $d \mid a$ and $d \mid b$.
2. If $c \mid a$ and $c \mid b$, then $c \leq d$.

Denote $d \overset{\text{def}}{=} \gcd(a, b)$.

Theorem 10.3.2. Let $a, b \in \mathbb{Z}$ be integers. Then, there exist integers $x, y \in \mathbb{Z}$ such that

$$\gcd(a, b) = ax + by$$

Proof. Define a set:

$$S \overset{\text{def}}{=} \{a \cdot u + b \cdot v \mid u, v \in \mathbb{Z},\ \ a \cdot u + b \cdot v > 0\}$$

Clearly, S is non-emptyset. Thus, S has a minimum element $d = \min S$ by Well-Ordering Principle.
Since $d \in S$, $d = ax + by$ for some $x, y \in \mathbb{Z}$. By the Division Algorithm, there exist $q, r \in \mathbb{Z}$ such that

$$a = d \cdot q + r \quad (0 \leq r < d)$$

That is,

$$r = a - d \cdot q = a - q \cdot (ax + by) = a \cdot (1 - qx) + b \cdot (-qy)$$

If $r \neq 0$, then $r \in S$ and $r < d$, this is contradiction with $d = \min S$. Thus, $r = 0$.
Similarly, $b = d \cdot q'$ for some $q' \in \mathbb{Z}$. In summary, d divides a and b.
Now, let c be an arbitrary positive common divisor of a and b. Then, c divides $ax + by = d$, thus $c \leq d$.
Consequently, $d = \gcd(a, b)$. □

Corollary 10.3.3. Let a, b be integers. For $d = \gcd(a, b)$,

$$\{a \cdot x + b \cdot y \mid x, y \in \mathbb{Z}\} = \{d \cdot k \mid k \in \mathbb{Z}\}$$

10.3.1 Extended Euclidean Algorithm

Lemma 10.3.4. Let a, b be integers.
If $a = b \cdot q + r$ with $0 \leq r < b$, then $\gcd(a,b) = \gcd(b,r)$.

Proof. Put $d = \gcd(a,b)$. Since $d \mid a$ and $d \mid b$, d divides $r = a - bq$. That is, d is common divisor of b and r. Meanwhile, let c be a divisor of b and r. Then, c divides $a = bq + r$, thus $c \leq d$. This implies $d = \gcd(b,r)$. □

Consider the given integers a, b. By the Division Algorithm, we can write:

$$
\begin{aligned}
a &= b \cdot q_0 + r_0 && (0 \leq r_0 < b) \\
b &= r_0 \cdot q_1 + r_1 && (0 \leq r_1 < r_0) \\
r_0 &= r_1 \cdot q_2 + r_2 && (0 \leq r_2 < r_1) \\
&\vdots \\
r_{k-2} &= r_{k-1} \cdot q_k + r_k && (0 \leq r_k < r_{k-1}) \\
&\vdots \\
r_{n-2} &= r_{n-1} \cdot q_n + r_n && (0 \leq r_n < r_{n-1}) \\
r_{n-1} &= r_n \cdot q_{n+1} && (r_{n+1} = 0)
\end{aligned}
$$

Since the Non-negative numbers Sequnece $\{r_k\}$ strictly decrease, r_{n+1} must be zero for some $n \in \mathbb{N}$.
Put $r_{-2} = a$ and $r_{-1} = b$. Since above Lemma,

$$\gcd(a,b) = \gcd(b,r_0) = \gcd(r_0,r_1) = \cdots = \gcd(r_{n-1},r_n) = \gcd(r_n,0) = r_n$$

This implies also for any $-2 \leq k \leq n$, r_k is multiple of r_n.
By Corollary of Bézout's Identity, for any $-2 \leq k \leq n$, there exists $x_k, y_k \in \mathbb{Z}$ such that

$$r_k = a \cdot x_k + b \cdot y_k \tag{$*$}$$

Since $r_{-2} = a$ and $r_{-1} = b$,

$$
\begin{aligned}
x_{-2} = 1,\ \ y_{-2} = 0 \\
x_{-1} = 0,\ \ y_{-1} = 1
\end{aligned}
$$

And, substituting $(*)$ to $r_{k-2} - r_{k-1} \cdot q_k = r_k$:

$$a \cdot x_k + b \cdot y_k = a \cdot x_{k-2} + b \cdot y_{k-2} - (a \cdot x_{k-1} + b \cdot y_{k-1}) \cdot q_k = a \cdot (x_{k-2} - x_{k-1} \cdot q_k) + b \cdot (y_{k-2} - y_{k-1} \cdot q_k)$$

Now, we obtain the equation: for all $0 \leq k \leq n$,

$$
\begin{aligned}
x_k &= x_{k-2} - x_{k-1} \cdot q_k \\
y_k &= y_{k-2} - y_{k-1} \cdot q_k
\end{aligned}
$$

Consequently, to find $x, y \in \mathbb{Z}$ such that $\gcd(a,b) = r_n = a \cdot x + b \cdot y = a \cdot x_n + b \cdot y_n$,
we simply calculate x_n, y_n inductively.

⌈Lemma. Given non-zero integers $a, b \in \mathbb{Z}$,

$$ax \equiv b \mod n \text{ has a solution if and only if } (a,n) \mid b.$$

In Particular, if $(a,n) \mid b$, then the equation has exactly (a,n) distinct roots.

Proof. $ax \equiv b \mod n$ for some $x \in \mathbb{Z} \Leftrightarrow n \mid ax-b. \Leftrightarrow \exists y \in \mathbb{Z},\ axy - b = ny.$

If $(a,n) \nmid b$ and $ax \equiv b \mod n$ has a solution, then

$$axy - ny = b \quad \text{but } (a,n) \nmid b.$$

But, by the Bézout identity, $axy - ny$ is multiple of (a,n), So contradiction.

So, $\Rightarrow$ Proved.⌋

Conversely, if $(a,n) \mid b$, then $b = (a,n) \cdot y$ for some $y \in \mathbb{Z}$.

So, $ax \equiv b \mod n \Leftrightarrow ax - (a,n)y$ is divided by n